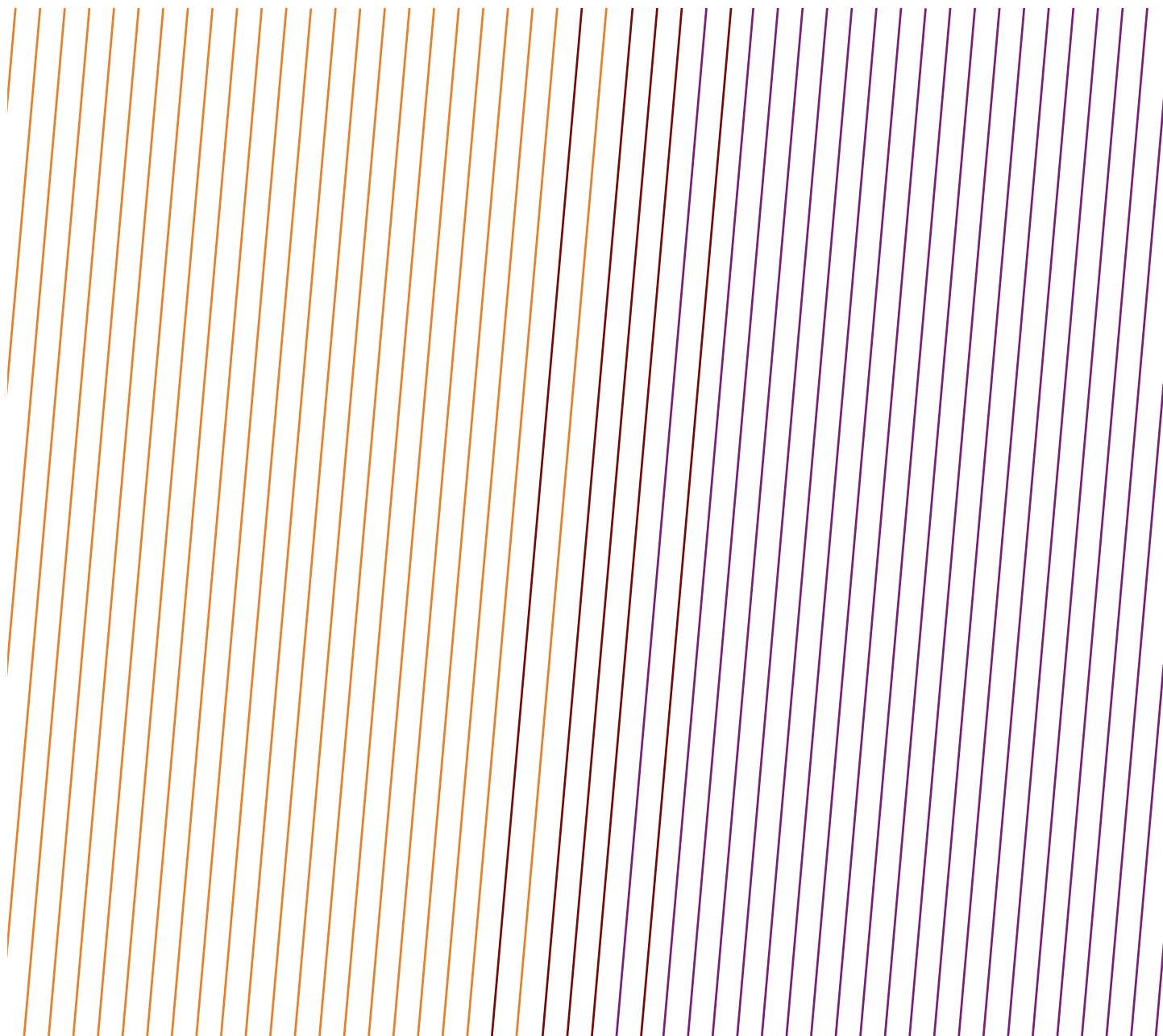


DATA SERIES

Safety performance indicators – 2021 data – Fatal incident reports



Acknowledgements

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This report was produced by the Safety Committee.

Feedback

IOGP welcomes feedback on our reports: publications@iogp.org

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DATA SERIES

Safety performance indicators – 2021 data – Fatal incident reports

Revision history

VERSION	DATE	AMENDMENTS
1.0	July 2022	First release

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AFRICA ONSHORE

DATE: Aug 16 2021

COUNTRY: Nigeria

FUNCTION: Construction

NUMBER OF DEATHS: 6

CAUSE: Assault or violent act,

ACTIVITY: Transport - Land

RULE: Unspecified

BODY PART: Unspecified

NATURE OF INJURY: Shot

Employer: Contractor	Occupation: Not available
Employer: Contractor	Occupation: Not available
Employer: Contractor	Occupation: Not available
Employer: Contractor	Occupation: Not available
Employer: Contractor	Occupation: Not available
Employer: Contractor	Occupation: Not available

NARRATIVE:

On 16th August, a convoy of contractor personnel enroute to the project site was caught in an ambush, resulting in 6 contractor employees fatally wounded and 7 individuals injured.

WHAT WENT WRONG:

- Act of Violence.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Implementation of Journey Management Plan
- Regular route surveys
- Security inspection

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PEOPLE (ACTS): Use of Protective Methods: Inadequate use of safety systems

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: Nov 1 2021

COUNTRY: Nigeria

FUNCTION: Construction

NUMBER OF DEATHS: 1

CAUSE: Falls from height

ACTIVITY: Construction, commissioning, decommissioning

RULE: Work at height

BODY PART: Respiratory system/internal organs

NATURE OF INJURY: Inhalation, poisoning, intoxication, asphyxiation, drowning

Employer: Contractor	Occupation: Not available
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NARRATIVE:

On November 1, 2021, a contractor personnel was fatally injured during a bulk pipeline repair.

AFRICA ONSHORE

WHAT WENT WRONG:

- Attorney Client Privileged Information.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Attorney Client Privileged Information.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate supervision

PROCESS (CONDITIONS): Organizational: Poor leadership/organizational culture

AFRICA OFFSHORE

No fatal incidents reported

ASIA / AUSTRALASIA ONSHORE

No fatal incidents reported

ASIA / AUSTRALASIA OFFSHORE

No fatal incidents reported

EUROPE ONSHORE

DATE: Oct 4 2021

COUNTRY: Romania

FUNCTION: Drilling

NUMBER OF DEATHS: 1

CAUSE: Caught in, under or between (excl. dropped objects)

ACTIVITY: Transport - Land

RULE: Driving

BODY PART: Respiratory system/internal organs

NATURE OF INJURY: Crushing

Employer: Contractor Occupation: Transportation Operator

NARRATIVE:

A driver employed by an upstream contractor had a fatal accident while driving the company trailer during a transport of equipment. The trailer he was driving rolled over on a public road and the cabin collapsed on him, leading to fatal injuries.

WHAT WENT WRONG:

Immediate/direct causes:

- Operating at improper speed

Root causes/ basic causes:

- Fatigue
- Improper decision making or lack of judgement
- Inadequate work standards/procedures:
- Inadequate /defective guards or protective barriers
- Inadequate /defective safety devices/warning systems
- Inadequate /defective tools/equipment/materials/products
- Poor risk awareness / risk assessment by the driver
- Alcohol policy not enforced and inconsistent and sporadic alcohol testing by Contractor
- Standard of Defensive Driving not followed by Contractor
- Journey Management: Inadequate use of IVMS (In Vehicle Monitoring System) by Contractor

Contributing factors/preconditions:

- Inadequate physical/physiological capability
- Inadequate leadership
- Inadequate tool/equipment/machinery/device
- Inadequate supervision/coaching
- Inadequate work/production standards
- Lack of knowledge
- Lack of skill

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Develop and roll out a risk awareness campaign
- Enforce alcohol policy by systematic testing and applying consequence management in all worksites

EUROPE ONSHORE

- Restart defensive driving training for all drivers with a 2-3 years interval refresher and with driver assessment
- Implement a transportation safety system as per standard requirements
- Ensure safe operation of all equipment by having a Preventative Maintenance System in place
- Verify implementation and effectiveness of transportation safety system as per standard

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PROCESS (CONDITIONS): Organizational: Inadequate communication

DATE: Jul 13 2021

COUNTRY: Romania

FUNCTION: Production

NUMBER OF DEATHS: 1

CAUSE: Caught in, under or between (excl. dropped objects)

ACTIVITY: Maintenance, inspection, testing

RULE: Line of fire

BODY PART: Respiratory system/internal organs

NATURE OF INJURY: Major/multiple system trauma

Employer: Contractor Occupation: Process/Equipment Operator

NARRATIVE:

At around 09:40, a trailer loaded with a backhoe arrived on the area and stopped on the right side of the road to unload the backhoe. During the preparation of unloading the backhoe, the trailer access ramp hit the backhoe operator. He was subsequently declared deceased.

WHAT WENT WRONG:

Immediate/direct causes:

- Failure to Identify Hazard
- Defective Tool/Equipment
- Incorrect / Inadequate Tool/Equipment

Root causes/ basic causes:

- Inadequate training/competence
- Inadequate work standards/procedures

Contributing factors/preconditions:

- Inadequate change
- Inadequate supply chain management
- Inadequate work/production standards
- Lack of skill

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Always secure the ramps with tension chains
- The Company will specify minimum requirements for transport fleet service standards beyond legal requirements
- Establish a development program for transportation service providers to meet the required service standards

EUROPE ONSHORE

- Change the emphasis from documents to the reality of behaviour and conditions in the field
- Develop an action plan to transform the mentality from repairs / maintenance to preventive maintenance.
- Review other contracts (other contractors for outsourced operations and projects) if there are similar issues.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate communication

EUROPE OFFSHORE

No fatal incidents reported

MIDDLE EAST ONSHORE

DATE: Jun 4 2021

COUNTRY: Kuwait

FUNCTION: Drilling

NUMBER OF DEATHS: 1

CAUSE: Caught in, under or between (excl. dropped objects)

ACTIVITY: Drilling, workover, well services

RULE: Line of fire

BODY PART: Respiratory system/internal organs

NATURE OF INJURY: Crushing

Employer: Contractor Occupation: Drilling/Well Servicing Operator

NARRATIVE:

At around 1:35 AM, two Roustabouts and a forklift with operator were assigned to prepare the casing joints at the catwalk prior to lifting them to rig floor. The Forklift was pushing the casing joints on pipe rack (12 Joints of 13 3/8" Casing) towards catwalk. As the casing joints were rolling, the Injured Person positioned himself in the line of fire between the catwalk and the casing which resulted in crush injury. After administering first aid on the rig, the Injured Person was transferred to Hospital using Rig Ambulance. Rig received information that Injured Person was declared deceased in the morning.

WHAT WENT WRONG:

1. Injured Person went into Line of Fire between catwalk & casing to pick up wooden plank.
2. Forklift Operator pushed casing close to catwalk.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

1. Line of Fire hazards to be avoided by drilling crew.
2. Establish clear communication by roustabouts and forklift operator while handling casings.
3. Strictly implement the controls highlighted in the Job Safety Analysis.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

MIDDLE EAST ONSHORE

DATE: Nov 18 2021

COUNTRY: Kuwait

FUNCTION: Production

NUMBER OF DEATHS: 1

CAUSE: Falls from height

ACTIVITY: Construction, commissioning, decommissioning

RULE: Work at height

BODY PART: Head (incl. ear, eye, mouth, etc.)

NATURE OF INJURY: Fracture

Employer: Contractor Occupation: Manual Labourer

NARRATIVE:

At around 13:00 hrs., six (6) workers (two at the ground, one at 4.4m, one at 6.9m, and two at 8.26m height) were involved in 'Scaffolding Erection' at GC (Water Knockout Pump Shelter). While one of the workers, at 8.26m height, was attempting to hook his Safety Harness, he lost his balance and fell, landing on a concrete surface and dying on the spot.

WHAT WENT WRONG:

- While attempting to hook his 'Safety Harness', he lost his balance and fell, landing on a concrete surface.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

1. All parties involved directly or indirectly in the work shall adhere to terms and conditions laid down in the contract document.
2. HSE Inspection / Audit plan for critical activities to be established at each asset.
3. Critical activity can be performed only under standalone valid Permit to Work specific to the job with dedicated supervision.
4. All hazards associated with the work to be thoroughly risk assessed and accordingly required control measures taken as a preventive action.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Failure to report/learn from events

MIDDLE EAST ONSHORE**DATE:** Mar 24 2021**COUNTRY:** UAE**FUNCTION:** Drilling**NUMBER OF DEATHS:** 1**CAUSE:** Falls from height**ACTIVITY:** Drilling, workover, well services**RULE:** Work at height**BODY PART:** Head (incl. ear, eye, mouth, etc.)**NATURE OF INJURY:** Unspecified

Employer: Company

Occupation: Drilling/Well Servicing Operator

NARRATIVE:

After rig release, drilling crew was involved in rig down activities for upcoming rig move. Major equipment on the rig floor was rigged down and lowered to the ground. Hydraulic jack pistons were installed into the substructure structural pad eyes (lifting point) as part of lowering substructure preparation. Driller side and off driller side rig floor plates above the jack pistons were removed. The crews left for a lunch break and upon arrival continued the work. While removing substructure's pins using a man lift, a crew member spotted the Driller inside the bottom box. He was unresponsive with signs of bleeding from nose and ears. Rig site medical response was activated and then he was evacuated to a hospital where he was pronounced dead.

WHAT WENT WRONG:

- Unprotected height (driller side and off driller side rig floor plates above the jack were removed, creating a gap).
- Lack of knowledge of hazards present (fall from the gap created by removing rig floor plates was not anticipated).
- Improper decision making or lack of judgement (a Contractor staff was at the rig floor unattended and access to the rig floor was not restricted while plates were removed).
- Inadequate work planning (there was a shortage of crew at the rig and a Contractor Driller was assigned to the rig).
- Inadequate technical design (the size of the gap created by the removal of rig plates was excessive).
- Inadequate management of change (the rig plates had been modified from original design - hinged deck hatch providing guard when open, to fully removable plates - without detailed review)
- Inadequate identification of worksite/job hazards (removal of floor plates was not addressed in job safety analysis (JSA) for lowering

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Limit the use of general permits or blanket permits.
- Review the existing JSA for lowering substructure to include opening floor plates.
- Introduce measures to formalize and communicate the buddying and supervisory requirements for each new-start or re-hire at site, regardless of their former experience.
- Define critical phases where major hazards exist during rig up and rig down operation to include controls like fall protection, barricading, restriction of access.

CAUSAL FACTORS:**PEOPLE (ACTS):** Use of Protective Methods: Inadequate use of safety systems**PEOPLE (ACTS):** Inattention/Lack of Awareness: Improper decision making or lack of judgment**PROCESS (CONDITIONS):** Organizational: Inadequate training/competence**PROCESS (CONDITIONS):** Organizational: Inadequate supervision

MIDDLE EAST OFFSHORE

DATE: Oct 21 2021

COUNTRY: UAE

FUNCTION: Drilling

NUMBER OF DEATHS: 1

CAUSE: Caught in, under or between (excl. dropped objects),

ACTIVITY: Lifting, crane, rigging, deck operations

RULE: Safe mechanical lifting

BODY PART: Neck/torso/spine

NATURE OF INJURY: Major/multiple system trauma

Employer: Contractor Occupation: Drilling/Well Servicing Operator

NARRATIVE:

During the preparation for an offshore jack-up drilling rig move, a deep well casing was being raised at STBD (Starboard) side using crane for securing and to remove a blind flange for installation in another deep well casing. During lifting activity, the roustabout found an Assistant Driller (IP) caught between the stopper handle and a fixed post. The roustabout informed the Banksman who informed the crane operator to stop and lower the deep well casing, to release the stopper handle.

The IP was taken to the rig clinic, the medic administered first aid, and a medivac request was initiated. HEMS team landed the rig at 10:10 hrs. and after checking the Injured Person, pronounced him dead.

WHAT WENT WRONG:

- Inadequate Audit/ Inspection/Monitoring: There was a gap in ensuring the availability of spare parts at the rig and no mechanism was in place to ensure critical spare parts are stocked.
- Inadequate Preventive Maintenance: There was no maintenance program for the stopper or stopper handle. All the stopper handles were removable and one of the handles was stuck at 45-degree angle, toward the fixed platform post, for some time.
- Inadequate Implementation of Standard/Procedure/Policies (PSP): A Permit to work (PTW) was issued for lifting the Deep well casing #2 but it did not include removal of blind flange from deep well casing #1; The activity was categorized as normal lift, although the crane operator had no visibility of the crane hook or load. The blind / complex lift categorization was not followed; In the available JSA, hazard and control measures (with respect to stopper handle, caught in between, blind flange removal, blind lift) were not addressed.
- Inadequate Supervisory Example: The work was supervised by a crane operator instead of the Barge Master, who was attending to other tasks in the Control Room.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Update the JSA of deep well casing retrieval / pump removal / blind flange procedure and associated risk.
- Survey all Offshore Drilling Jack up units for deep well pump system and ensure it is safe to operate in compliance with the Original Equipment Manufacturer (OEM) manual.
- Initiate engineering study to review the deep well casing design, stopper mechanism and work layout.
- Ensure all safety requirements for deep Well casing retrieval must be in line with blind lift (non-routine complicated) requirements as per Company Offshore LIMS.
- Issue a new standing instruction to all the CPLD / Barge Master / crane operator / Banksman / Rigger to reinforce the responsibility requirements and compliance requirements as per Company Offshore LIMS.
- Drilling Contractor to study installation of CCTV that can capture all rig locations with recording facility.

MIDDLE EAST OFFSHORE

- Ensure availability of adequate spare parts as per the OEM Manual and equipment integrity and maintenance.
- Review and update HSE terms of contracts and develop a clear RACI chart covering different services.

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Organizational: Inadequate communication

NORTH AMERICA ONSHORE

No fatal incidents reported

NORTH AMERICA OFFSHORE

DATE: Jan 24 2021

COUNTRY: USA

FUNCTION: Drilling

NUMBER OF DEATHS: 1

CAUSE: Caught in, under or between (excl. dropped objects)

ACTIVITY: Maintenance, inspection, testing

RULE: Line of fire

BODY PART: Respiratory system/internal organs

NATURE OF INJURY: Crushing

Employer: Contractor Occupation: Foreman, Supervisor

NARRATIVE:

At around 1330, two Asset Management company employees were performing loose lifting gear inspections as per the Contractor performance management system (PM). This task involves visually inspecting wire rope slings and shackles, recording the serial or batch numbers and painting the new colour code on the slings (in accordance with the Contractor company's Lifting Manual). This service is provided to Contractor rigs by the Asset Management company. The IP (Asset Management company supervisor) was attempting to remove a wire sling that was wrapped around five steel plates for

NORTH AMERICA OFFSHORE

inspection. The plates were supported against a stanchion post on the starboard aft riser deck. As a result of the storage arrangement and securing method, the sling was pinched between the plates and stanchion, preventing it from being removed for inspection. The IP summoned his co-worker to assist in removal of the sling and instructed him to loosen the ratchet strap that was securing the plates against the stanchion, presumably to allow the plates to move away from the stanchion and free the pinched sling. The IP attempted to physically support the plates in a near vertical position by positioning his back to the plates, so the sling could be removed. After loosening the ratchet strap, the hook came free and the plates fell away from the stanchion post. As the plates fell, the IP was unable to support the plates due to the weight ($\pm$ 4,000 pounds) and was unable to escape. The plates continued to tilt until the IP was pinned between the plates and an adjacent stationary steel frame. Several people ran to the scene immediately upon hearing calls for help, but the plates were too heavy for them to lift. The IP suffered fatal injuries as a result.

WHAT WENT WRONG:

- The unauthorized operation of ratchet straps has never been identified as a significant likelihood.
- PM provider did not direct or control the lifting gear inspection service provider.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Review arrangements and marking for Restricted Access/Safe Areas. Ensure such areas consider line of fire and proximity to danger, as well as clear access into and out of the area.
- Review all moving plant and machinery for similar caught between and proximity to danger hazards. Ensure Controlled/Rig Procedures and Risk Assessments reference the presence of such hazards.
- Review the presence and positioning of manually activated emergency stop buttons on moving plant and machinery.
- Consider the use of proximity barriers to prevent personnel access to areas where plant and machinery is moving.
- Consider audible and visual alarms for plant and machinery that moves.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

RUSSIA & CENTRAL ASIA ONSHORE

DATE: Oct 10 2021

COUNTRY: Kazakhstan

FUNCTION: Production

NUMBER OF DEATHS: 1

CAUSE: Exposure electrical,

ACTIVITY: Maintenance, inspection, testing

RULE: Energy isolation

BODY PART: Respiratory system/internal organs

NATURE OF INJURY: Electric shock

Employer: Company Occupation: Maintenance, Craftsman

NARRATIVE:

During maintenance works, an Electrical Technician received a fatal electrical shock whilst performing a 10KV isolation of a transformer that fed controls and distribution to a wellsite beam pump. The fatally injured person touched a live busbar of the incoming 10KV feed.

WHAT WENT WRONG:

- The isolation point of the 3-phase supply was at a line disconnecter for outdoor installations; a pylon mounted overhead isolator designed to segregate overhead 3-phase.
- Although the 'isolate' and 'ground' levers of the pylon were operated and locked off, a connector for one of the phases failed to open as intended. The mechanism that opens the overhead knife contacts is a mechanical linkage. A nut which secures the linkage
- to one of the knives was loose/missing, resulting in the knife staying closed when the isolating lever was operated. Analysis shows that the nut had been loose prior to the incident.
- The 10KV feed had not been effectively isolated and verified.
- The Electrical Technician wore rubber gloves whilst operating the isolating levers, but those gloves were not rated for High Voltage (HV) protection.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Ensure local electrical isolating procedures detail safe methods of testing and verification, specifying the correct PPE to be used.
- Ensure that local, equipment specific isolating procedures are clear on the critical points of verification; and that wherever additional physical checks are required prior to Zero Volt Checking, that the Supervisor verifies 'safe to proceed'.
- Ensure that the isolating authority appointment processes are adequately robust in their demonstration of electrical competence for performing, verifying, and testing of High Voltage isolations.
- Never perform work on HV systems under personal isolation. Personal isolation is only applicable to low-risk work, and therefore not allowed on HV systems.

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Fatigue

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

RUSSIA & CENTRAL ASIA ONSHORE

DATE: Jun 12 2021

COUNTRY: Kazakhstan

FUNCTION: Construction

NUMBER OF DEATHS: 1

CAUSE: Struck by (not dropped object)

ACTIVITY: Construction, commissioning, decommissioning

RULE: Line of fire

BODY PART: Head (incl. ear, eye, mouth, etc.)

NATURE OF INJURY: Crushing

Employer: Contractor Occupation: Not available

NARRATIVE:

The crew consisted of several dump trucks loading and transporting sand from the stockpile site to worksite adjacent to the construction area. The fatality occurred when the individual was struck by a dump truck manoeuvring in the open stockpile area.

WHAT WENT WRONG:

- Attorney Client Privilege Information.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Attorney Client Privilege Information.

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgment

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

RUSSIA & CENTRAL ASIA OFFSHORE

No fatal incidents reported

SOUTH & CENTRAL AMERICA ONSHORE

DATE: Jun 7 2021

COUNTRY: Argentina

FUNCTION: Drilling

NUMBER OF DEATHS: 1

CAUSE: Caught in, under or between (excl. dropped objects)

ACTIVITY: Maintenance, inspection, testing

RULE: Line of fire

BODY PART: Respiratory system/internal organs

NATURE OF INJURY: Major/multiple system trauma

NARRATIVE:

While inspecting the rear axle of the telehandler, the technician enlisted the assistance of the Welder Supervisor to steer the equipment. When performing the requested turns, the technician was trapped between the wheel and the rear of the equipment.

WHAT WENT WRONG:

- None of the interviewed people knew about the sub-contractor Technician's arrival.
- The Control of Work/Risk Analysis was not performed.
- The equipment was turned on by unauthorized personnel
- Wheel moved as mechanical lock was not attached to rear differential

Planning failure:

- Define to carry out on-site transmission bar welding.
- Unsupervised task

Location Access:

- The telehandler was left out of the control zone.

Maintenance management:

- The criteria of which jobs can be done in the field and which in the workshop is not defined.
- There is no technical reference to define maintenance work.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Change the entry process to a location with modification of the location of the barrier and adaptation of the entry form, stating that the person received the induction.
- Activate the locking system that only allows the start of the telescopic handler with certified personnel.
- Incorporate the requirement of a certified Telescopic Handler operator / maintainer.
- Maintenance management: Redefine the process and scope of the management of the Telescopic Handler aligned to the Rigs (Management of major deviations, safety devices, Field and workshop tasks, Technical Competencies, R&R).

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

SOUTH & CENTRAL AMERICA ONSHORE

DATE: Apr 23 2021

COUNTRY: Argentina

FUNCTION: Drilling

NUMBER OF DEATHS: 1

CAUSE: Struck by (not dropped object)

ACTIVITY: Lifting, crane, rigging, deck operations

RULE: Safe mechanical lifting

BODY PART: Leg/knee/hip

NATURE OF INJURY: Fracture

NARRATIVE:

While performing a kelly lifting operation (weight: 2182 kg, length: 13 m) from a truck to a pipe ramp using a lorry crane, the load became unstable due to the loose bushing and it began to sway. In that moment the load hit the worker's lower limbs, fracturing both legs and resulting in a fatal injury.

WHAT WENT WRONG:

Risk Assessments were inadequate:

- Unstable load was not identified as a potential Hazard.
- Kelly Bushing was not fixed while performing the movement.

Some critical Processes and procedures were not followed:

- The lifting plan was not requested.
- Sub-contractor was not supervised during the lifting operations.
- Control of Work procedure was not followed.
- Inspection did not detect previous deviations.
- Lifting operator certification carried out by a body not approved by OA.
- Contractor did not comply with bridging document.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Define the proper way of loading, unloading and transporting Swivel, Kelly, and bushing.
- Ensure HSE requirements in operations finished
- Review and update Bridging document for sub-contractors.
- Review of the Contractor Resources Qualification procedure, incorporation of acceptance criteria for operators and lifting equipment accreditation entities.
- Implement discipline program.
- Train all team leaders and Shifts of the Completion teams in control of work, load lifting, work permits and golden rules.

CAUSAL FACTORS:

No Causal Factors Allocated

SOUTH & CENTRAL AMERICA OFFSHORE

DATE: Jan 31 2021

COUNTRY: Brazil

FUNCTION: Production

NUMBER OF DEATHS: 1

CAUSE: Falls from height

ACTIVITY: Maintenance, inspection, testing

RULE: Work at height

BODY PART: Unspecified

NATURE OF INJURY: Major/multiple system trauma

Employer: Contractor Occupation: Maintenance, Craftsman

NARRATIVE:

During inspection activity and UTM (thickness measurement) inside the 7S cargo tank, a worker fell approximately 18 metres.

WHAT WENT WRONG:

- Poor job planning
- Inadequate supervision
- Inadequate procedure
- Risk management failure
- PTW failure

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Review the Rope Access Work procedure.
- Create a management guide for thermal stress, establishing guidelines to be followed during the planning and execution phases of activities that may be influenced by thermal discomfort.
- Review the Work Permit procedure.

CAUSAL FACTORS:

No Causal Factors Allocated

DATE: Sep 25 2021

COUNTRY: Brazil

FUNCTION: Production

NUMBER OF DEATHS: 1

CAUSE: Water related, drowning

ACTIVITY: Transport - Water, incl. marine activity

RULE: Safe mechanical lifting

BODY PART: Respiratory system/internal organs

NATURE OF INJURY: Inhalation, poisoning, intoxication, asphyxiation, drowning

Employer: Contractor Occupation: Other

NARRATIVE:

An Oil Spill Recovery Vessel (OSRV) was carrying out a routine simulation exercise by testing its recovery spill oil equipment. At the beginning of the exercise, while launching a workboat into the water with three crew members on board, using the vessel's crane, the boat fell into the water.

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Two crew members got off the boat with no injuries. A rescue team was used to take out the remaining crew member on the workboat, but they found him already dead.

WHAT WENT WRONG:

- Unknown or not accessible procedure;
- Inadequate or non-existent risk assessment;
- Inadequate design requirements.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Vessels leadership must ensure that all crane equipment that transports people from the installation, has competent personnel in maintenance and inspection.
- Competent personnel must verify that components of steel cables have been executed/assembled according to ISO 17558 for open sockets and ASME B30.26-2015 (item 3.1.4) for wedge sockets.
- Vessel leadership must verify that the Risk Analysis of boats / lifeboats / transshipment baskets have identification of risks and mitigations in the event of an abrupt fall with people on board and the lifeboat remaining in an overturned position.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

For the full analysis of 2021 results,
as well as all fatal incident reports
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<https://data.iogp.org>

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